Order-to-Cash (O2C) — Complete Sales Cycle using SAP SD

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1. Title

Implementation and Simulation of Order-to-Cash (O2C) Business Process using SAP SD with Web-Based Validation

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2. Problem Statement

In modern enterprises, managing the complete sales lifecycle—from receiving customer orders to collecting payments—is complex and prone to inefficiencies when handled through disconnected systems.

Organizations face challenges such as:

- Lack of real-time inventory visibility
- Delays in order processing
- Errors in billing and invoicing
- Poor integration between sales and finance

The objective of this project is to design and simulate a fully integrated **Order-to-Cash (O2C)** process using SAP SD, ensuring seamless coordination between Sales (SD), Materials Management (MM), and Financial Accounting (FI).

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3. Company Scenario (Fictitious Data)

- **Company:** Vertex Global Pvt. Ltd.
- **Industry:** Electronics Distribution
- **Customer:** Bright Electronics
- **Material:** LED-TV-1000
- **Order Quantity:** 100 Units

- **Unit Price:** ₹25,000
- **Total Order Value:** ₹25,00,000

4. Solution / Features

The project implements a complete O2C cycle with:

- Customer Master Data Management
- Sales Order Processing
- Availability Check (ATP)
- Delivery & Shipping
- Billing & Invoice Generation
- Payment Processing
- Integration with FI and MM
- Web-based simulation for validation

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5. Step-by-Step O2C Process

Step 1: Customer Master Creation

- **T-Code:** XD01
- **Tables:** KNA1, KNB1, KNVV
- Stores customer details

Step 2: Inquiry

- **T-Code:** VA11
- Pre-sales document
- No financial impact

Step 3: Quotation

- **T-Code:** VA21
- Pricing and validity maintained

Step 4: Sales Order

- **T-Code:** VA01
- **Tables:** VBAK (Header), VBAP (Item)

Impact:

- Demand created
- ATP check triggered

Step 5: Delivery

- **T-Code:** VL01N
- **Tables:** LIKP, LIPS

Step 6: Goods Issue

- **T-Code:** VL02N
- **Movement Type:** 601

Tables Updated:

- MKPF, MSEG, MARD

Accounting Entry:

- Dr Cost of Goods Sold
- Cr Inventory

Step 7: Billing

- **T-Code:** VF01

Tables Updated:

- VBRK, VBRP
- BKPF, BSEG

Accounting Entry:

- Dr Customer
- Cr Revenue

Step 8: Payment

- **T-Code:** F-28

Accounting Entry:

- Dr Bank
- Cr Customer

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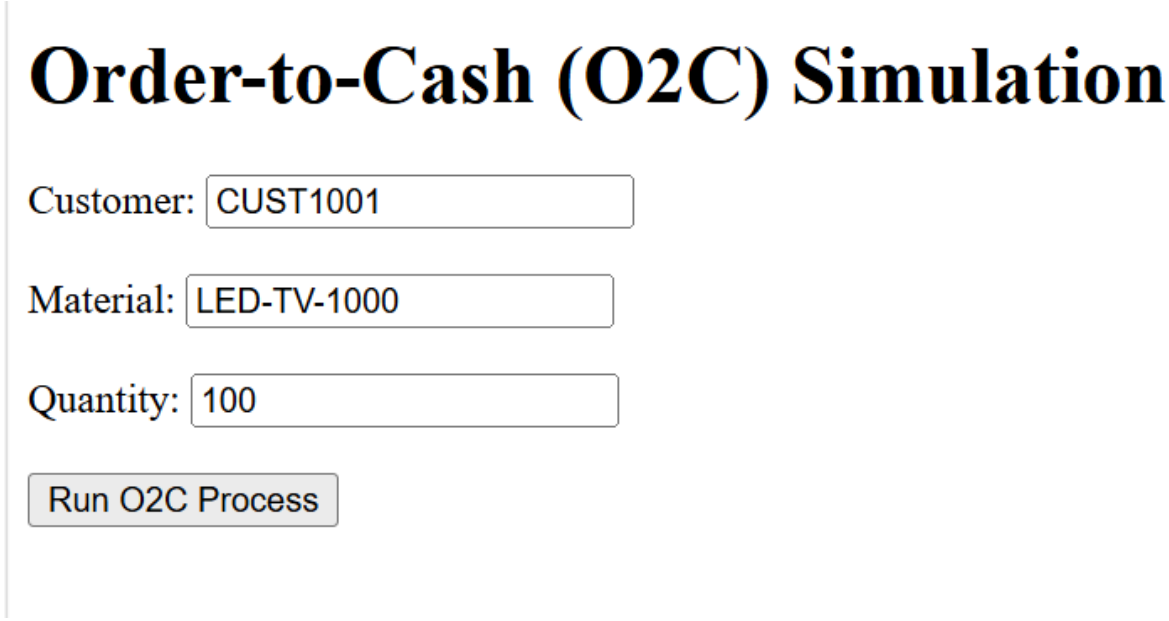
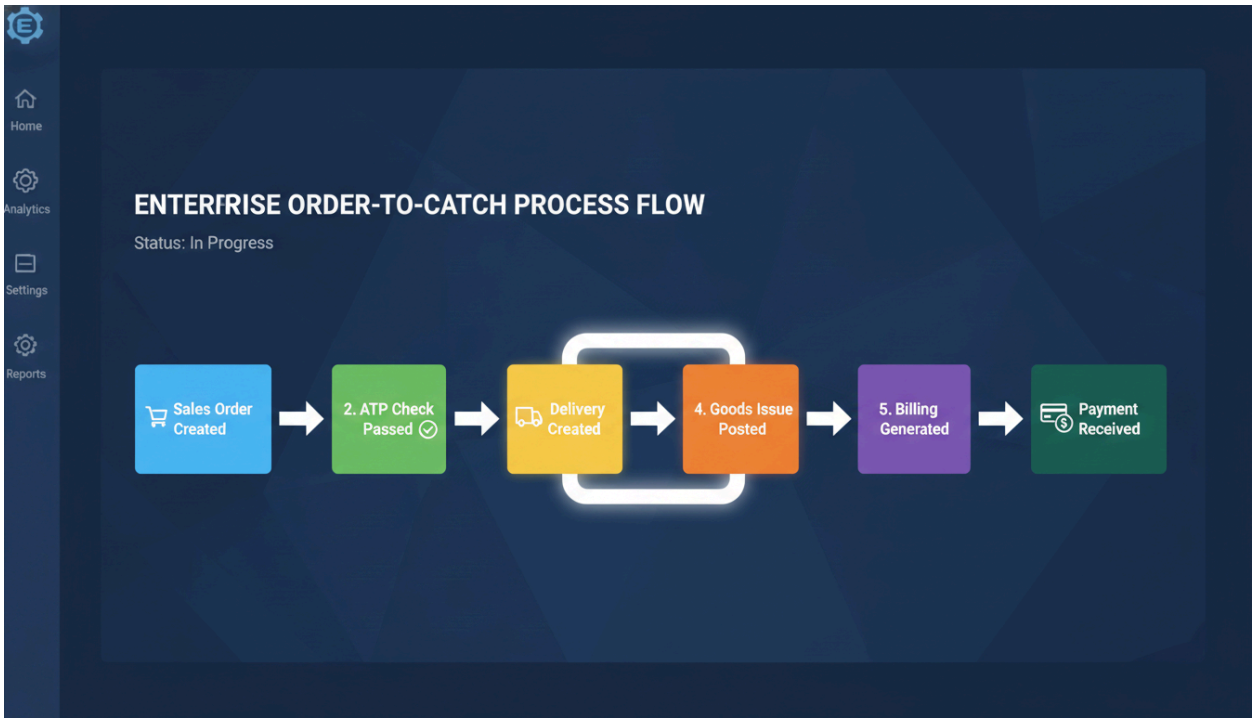
6. Process Flow Summary Table

Step	Document	T-Code	Tables Updated	Accounting Impact
1	Customer Master	XD01	KNA1, KNB1, KNVV	None
2	Inquiry	VA11	VBAK, VBAP	None
3	Quotation	VA21	VBAK, VBAP	None
4	Sales Order	VA01	VBAK, VBAP	Demand Created
5	Delivery	VL01N	LIKP, LIPS	None
6	Goods Issue	VL02N	MKPF, MSEG, MARD	Dr COGS / Cr Inventory
7	Billing	VF01	VBRK, VBRP, BKPF, BSEG	Dr Customer / Cr Revenue
8	Payment	F-28	BSID, BSAK	Dr Bank / Cr Customer

7. Screenshots





8. Tech Stack

SAP System

- SAP ERP (ECC / S/4HANA)
- Modules: SD, MM, FI

Development

- Python (Flask Web App)
- HTML (Frontend UI)

Tools

- SAP GUI
- VS Code
- GitHub

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9. Unique Points

- End-to-end O2C cycle implementation
- Real SAP tables and T-code mapping
- Integration across SD, MM, FI
- Simulation of ATP check
- Web application validating SAP process logic
- Combination of functional + technical approach

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10. Web Application Justification

A mini web application was developed using Flask to simulate the O2C process.

The system replicates:

- Sales Order Creation
- Availability Check (ATP)
- Delivery Processing
- Billing and Payment

This validates the SAP process by demonstrating:

- Logical flow of transactions
- Decision-making (stock validation)
- Financial postings

Thus, the project is not only theoretical but also practically implemented.

11. Future Improvements

- Integration with SAP Fiori UI
- Real-time dashboard and analytics
- Credit management system
- Cloud deployment (AWS / SAP BTP)
- AI-based demand forecasting

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12. Conclusion

The Order-to-Cash process demonstrates how SAP integrates multiple business functions into a unified system. This project successfully simulates a real-world sales cycle and validates it through both SAP concepts and a working web application.